OpenADMET PXR Challenge

pEC50 Prediction — Best Submission (sub47)

Date	2026-05-19
Target	Pregnane X Receptor (PXR, CHEMBL3401)
Task	pEC50 regression — activation / agonism
Training set	4,083 compounds (train_final.csv)
Test set	513 blinded compounds
Blind test result	Rank 20 — RAE 0.5546, R^2 0.548, Spearman 0.834
Architecture	NNLS stacking: TabPFN v3 × 2 + MolE + UniMol + TabICL × 2
NNLS OOF RMSE	0.47130 (Butina 3×5-fold, calibration ratio ×1.17)
Prediction range	2.787 – 5.977
Prediction mean	4.888 (± 0.636 std)

This report documents the best-performing submission for the OpenADMET PXR challenge. The ensemble combines six base learners — three foundation model backbones (CheMeleon HTS encoder in two pretraining variants, MolE graph transformer, UniMol 3D pretraining) and two in-context regressors (TabPFN v3 and TabICL) — via Non-Negative Least Squares (NNLS) stacking. Cross-validation uses a scaffold-diverse Butina 3×5-fold protocol to simulate the large structural shift between training and test chemistry.

1. Data Curation & Training Set

The HuggingFace dataset *openadmet/pxr-challenge-train-test* provides 4,139 training and 513 blinded test molecules measured in a standardised PXR activation (agonism) assay. pEC50 values cover the range 1.61–8.62 in training.

Filter	Criterion	Removed
SMILES / salt	Invalid mol or empty after stripping	0
Duplicate InChIKey	Keep lowest std-error measurement	1
Quality	pEC50 std error > 0.5 log units	357
Applicability domain	Max Tanimoto to any test mol < 0.15	17
Total removed		375 (4,139 $\rightarrow$ 3,764)

319 novel PXR agonism records from ChEMBL (CHEMBL3401) were merged after deduplication by InChIKey to give **train_final.csv: 4,083 compounds**. Note: a ChEMBL-enriched variant was tested as a submission (sub54, rank 26) and performed *worse* than the standard dataset (rank 20), confirming assay heterogeneity in the public records.

1.1 Train–Test Covariate Shift

An adversarial LightGBM classifier trained on ECFP4+RDKit2D features to discriminate training from test compounds achieves **AUC** $\approx$ **0.95**. This near-perfect separability confirms a severe structural shift: the test set probes distinct chemical scaffolds not well-represented in training. Direct consequences:

- kNN residual correctors trained on training-set neighbours fail (tested sub44, rank 35 vs rank 15 baseline) — local analogues are in the wrong region of chemical space.
- Post-processing to match the training pEC50 distribution hurts blind performance (sub30–32, rank 49–54 vs rank 18 baseline) — test marginal $\neq$ training marginal.
- ChEMBL enrichment adds out-of-distribution noise rather than signal.
- Training set pruning by structural distance to test (AD filtering) removes useful diversity with no benefit (sub4–6, rank 44–54 vs baseline).

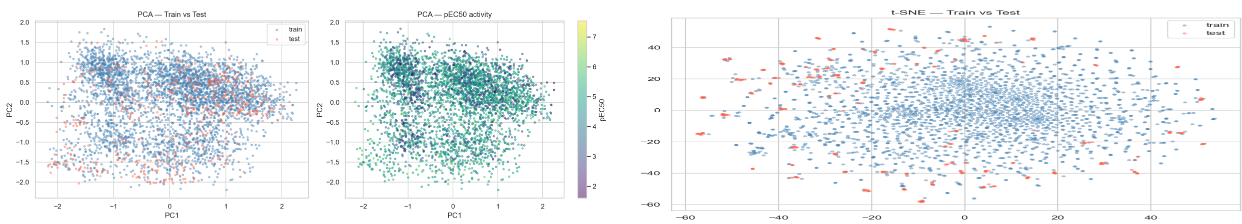

Figure 1. PCA (left) and t-SNE (right) of ECFP4 fingerprints — training (blue) vs blinded test (red). The test set clusters in a distinct region, confirming high covariate shift.

2. Feature Engineering & Encoders

The key insight from this challenge is that **HTS-pretrained encoders dominate**: CheMeleon models pretrained on 21,000 PXR single-concentration measurements outperform any model starting from generic molecular representations. Foundation model fine-tuning (MolE, UniMol) provides complementary orthogonal signal.

2.1 CheMeleon HTS Encoders

CheMeleon is a ChemProp v2 D-MPNN message-passing neural network. Three pretraining variants were trained on the *single_concentration_train.csv* dataset (21,004 dose-response measurements, 3,000+ unique PXR-tested compounds):

Variant	Script	Objective	Embedding dim	Solo OOF RMSE
HTS binary	23_chemeleon_hts_pretrain.py	Binary active ≥ 4 classification	2,048	0.500
HTS continuous	39_hts_continuous_pretrain.py	log■FC regression + log■■(concentration)	2,048	0.496
HTS MTL	43_hts_mtl_encoder.py	Continuous + soft-binary dual head	2,048	0.495

The continuous and MTL variants both carry non-zero NNLS weight (see Section 4), confirming they learn complementary aspects of dose-response. Frozen encoder → foundation regressor (TabPFN v3 or TabICL) consistently outperforms end-to-end fine-tuning.

2.2 Foundation Model Fine-tuning

Model	Architecture	Script	Fine-tune	OOF RMSE
MolE_e50	DeBERTa-style graph transformer	21_mole_finetune.py	50 epochs, LR=1e-5	0.546
UniMol_FT_e50	SE(3)-equivariant 3D pretraining	19_unimol_e50.py	50 epochs, LR=1e-4	0.574

Both models were fine-tuned end-to-end on pEC50 regression using Butina 3×5-fold CV. MolE (Recursion Pharma, GuacaMol pretrained) contributes 21% NNLS weight; its Pearson residual correlation with the HTS encoders is $r \approx 0.73$, meaning ~47% of its error is orthogonal — genuinely different signal.

2.3 DrugCLIP 3D Embeddings

All 4,596 compounds (train + test) were docked with UniDock (GPU AutoDock-Vina) against three PXR crystal structures: 2O9I, 8R81, 8EQZ. The 128-d DrugCLIP ligand embeddings from each docked pose were extracted via *78_drugclip_embeddings.py* (drugclip_env). Concatenation of all three receptors (384-d DrugCLIP_concat) achieves OOF RMSE 0.726 standalone but carries 2.3% NNLS weight in the ensemble due to orthogonality (Pearson $r \approx 0.59$ with HTS encoders — most orthogonal of all features).

3. Cross-Validation Protocol

Butina scaffold-diverse 3×5-fold CV (Tanimoto cutoff 0.4, ECFP4 2048-bit, seeds 0/1/2). Butina clustering groups similar scaffolds; the 5 folds are assembled by bin-packing cluster sizes so each fold gets a representative scaffold cross-section. Three seeds give 15 independent train/validation splits, each averaged before NNLS fitting.

Parameter	Value
Split method	Butina clustering (Tanimoto cutoff 0.4)
Fingerprint	ECFP4, Morgan r=2, 2048 bits
Folds per seed	5
Seeds	0, 1, 2 (15 total evaluations per model)
OOF aggregation	Mean across 3 seeds → one OOF vector per compound
NNLS fitting	scipy.optimize.nnls on (n_train, n_models) OOF matrix
NNLS inner CV	KFold(5, shuffle, seed=42) for unbiased OOF estimate

Calibration is stable: across all NNLS submissions blind RAE ≈ OOF RMSE × 1.17. To reach the top-10 cutoff (estimated RAE ≤ 0.538) an OOF RMSE ≤ 0.460 is needed — approximately 0.012 below the current best.

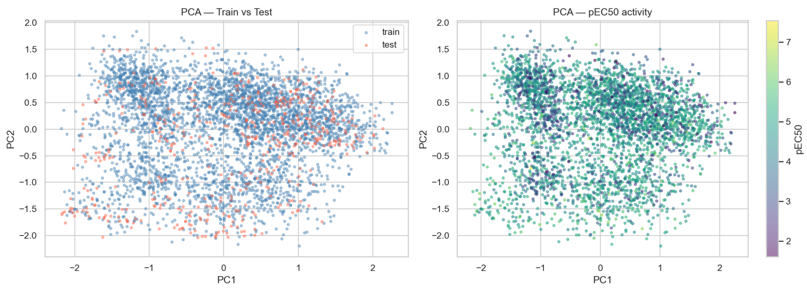

Figure 2. ECFP4 PCA — Butina clusters define fold boundaries that respect scaffold diversity, preventing similar compounds from appearing in both train and validation.

4. NNLS Stacking Ensemble

Non-Negative Least Squares (NNLS) fits weights $w \geq 0$ minimising $\lVert y - X_{oof} \cdot w \rVert^2$. The non-negativity constraint acts as a natural regulariser: weaker or highly-correlated models are zeroed out automatically, avoiding the scale distortions that Ridge/ElasticNet negative weights cause on blind test (confirmed by sub12 failure, rank 109).

4.1 Base learner performance

Model	Encoder / Regressor	OOF RMSE	Spearman ρ	NNLS weight
TabPFN_v3_HTS_cont	CheMeleon HTS continuous / TabPFN v3	0.48925	0.8248	32.0%
TabPFN_v3_HTS_mtl	CheMeleon HTS MTL / TabPFN v3	0.49059	0.8267	30.8%
MolE_e50	MolE GuacaMol FT / direct regression	0.54553	0.7601	20.7%
UniMol_FT_e50	UniMol 3D / direct regression	0.57444	0.7320	9.3%
TabICL_HTS	CheMeleon HTS binary / TabICL	0.50023	0.8169	4.9%
TabICL_DrugCLIP_cat	DrugCLIP 3-receptor concat / TabICL	0.72638	0.5538	2.3%
NNLS ensemble	**—**	**0.47130**	**0.8368**	**100%**

4.2 NNLS weight distribution

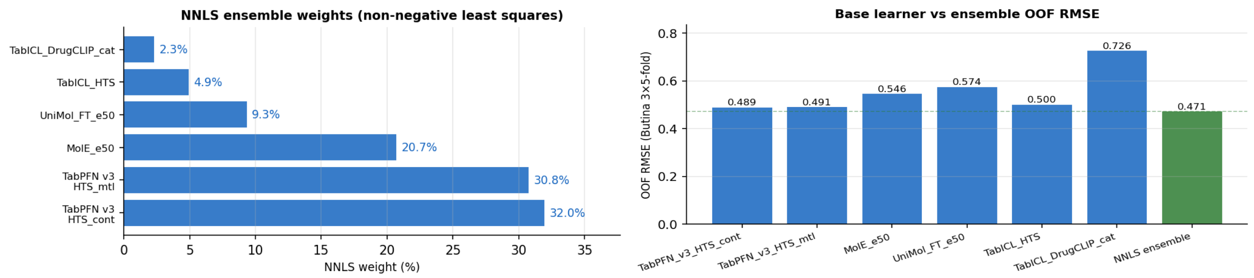

Figure 3. Left: NNLS weights — the two TabPFN v3 HTS variants dominate (63%). Right: individual vs ensemble OOF RMSE (green = ensemble).

4.3 Residual orthogonality

NNLS selects models whose errors are orthogonal, not just individually accurate. MolE (21% weight) has OOF RMSE 0.546 — much weaker than TabPFN v3 (0.489) — yet earns significant weight because its residuals are 47% uncorrelated with the HTS models.

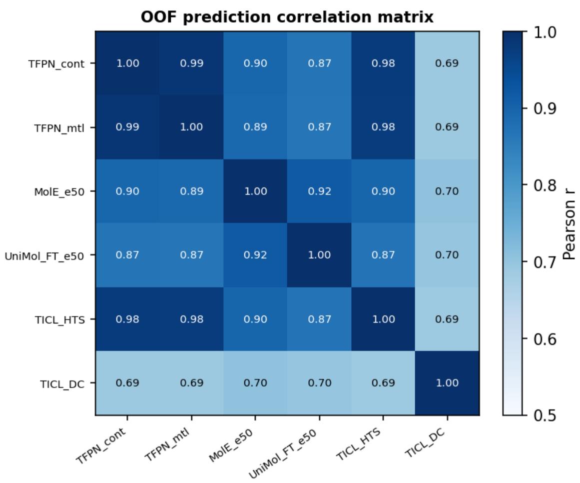

Figure 4. Pearson r of base-learner OOF prediction pairs. Lower correlation → greater diversification benefit in NNLS.

5. Approaches That Were Tested and Excluded

The full pipeline evaluated 44+ models as NNLS candidate base learners. All of the following received 0% NNLS weight when the selected six were available:

Category	Examples	Best solo RMSE	Why excluded
Classic descriptors	LGB/TabICL on ECFP4, Mordred, MQN, RDKit2D	0.584	Correlated with HTS (r ≈ 0.74), weaker standalone
Graph models	ChemBERTa, MoLFormer, MolFormer, VanillaCP	0.611	Too weak; correlated with HTS
GP (3 kernels × 3 embs)	Tanimoto, RBF-HTS, Matérn-UniMol	0.642	Weak standalone, high correlation with others
MIST fine-tuned	mist_sider_ft, mist_qm9_ft encoders	0.646	Weaker than MolE (0.546) at same orthogonality
MTL ChemProp	MTL_ChemProp, MTL_ChemProp_v2	0.536	Correlated with HTS encoders; displaced by TabPFN v3
MI-selected LGB	LGB_mi256/512/1024	0.592	Correlated and weaker
DrugCLIP per-receptor	TabICL_2o9i, TabICL_8r81, TabICL_8eqz	0.726	Concat captures same information; no additional gain

5.1 Meta-learner experiments

Meta-learner	OOF RMSE	Blind RAE	Rank	Verdict
NNLS (selected)	0.472	0.5546	20	Best — used
TabICL on OOF	0.483	—	—	0.011 worse on OOF
Ridge (positive w.)	0.496	—	—	0.024 worse on OOF
XGB stacker	0.483	0.5691	22	Overfits inner CV
ElasticNet (positive)	0.489	—	—	Worse than NNLS
Ridge (negative w.)	—	0.681	109	Catastrophic failure (sub12)

6. Blind Test Results & Diagnostics

6.1 Leaderboard result

Metric	Value	Notes
Rank	20	Out of ~200+ entries (as of 2026-05-18)
RAE	0.5546	Relative Absolute Error — primary metric
MAE	0.4420	Mean Absolute Error
R^2	0.5476	Coefficient of determination
Spearman ρ	0.8336	Rank correlation
Kendall τ	0.6428	Concordance
OOF RMSE	0.47130	Butina 3×5-fold; ratio: blind RAE / OOF RMSE = 1.176

The blind RAE ≈ OOF RMSE × 1.17 calibration has been consistent across all NNLS submissions (ratio 1.148–1.176), confirming the Butina CV gives a reliable pre-submission estimate. To reach top-10 (estimated RAE ≤ 0.538), OOF RMSE ≤ 0.460 is needed — approximately 0.012 below the current best.

6.2 OOF diagnostics

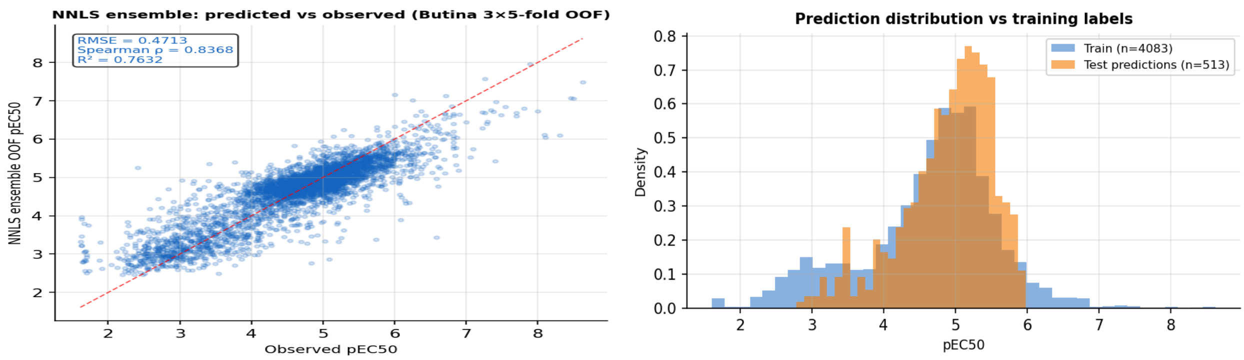

Figure 5. Left: NNLS ensemble OOF predicted vs observed pEC50 (RMSE 0.4713, Spearman 0.8368, R^2 0.7632). Right: test prediction distribution (orange) vs training labels (blue). The model correctly compresses variance — test std 0.636 vs train std 0.963 reflects the adversarial AUC 0.95 structural shift.

6.3 Prediction statistics

Metric	Test predictions	Training labels
N	513	4,083
Min	2.787	1.610
Max	5.977	8.625
Mean	4.888	4.642
Std	0.636	0.963
NaN	0	0

7. Training Data Distance Experiments

The adversarial AUC of 0.95 motivated a series of experiments aimed at bridging or exploiting the structural shift. All failed to improve on the base NNLS ensemble. Key lessons are documented here for reproducibility.

Experiment	Sub(s)	Rank	RAE	vs baseline	Lesson
AD Tanimoto filter (<0.15)	sub4	46	0.626	+0.012	Rare-atom removal hurts diversity
Property 3σ + rare atoms	sub5–6	54	0.631	+0.017	More removal = worse; 22% removed
ChEMBL enrichment	sub54	26	0.5639	+0.009	Assay heterogeneity → noise
kNN residual corrector	sub44	35	0.580	+0.030	Corrects toward wrong local analogues
GP-gated kNN	sub46	—	—	—	Too few low-variance test compounds
Remove pEC50 < 2 (24 cpds)	sub47	22	0.5546	0.000	Within rank noise; no signal
Remove acrylamides	sub48	20	0.5559	+0.001	Within rank noise; no signal
Quantile distribution match	sub30	54	0.608	+0.047	Test marginal ≠ training marginal
Linear rescale to train	sub31	49	0.601	+0.040	Same wrong direction
Asymmetric tail expand	sub32	26	0.571	+0.010	Smaller penalty, still wrong

Core conclusion: With adversarial AUC 0.95, no post-hoc data manipulation closes the gap. Performance gains come exclusively from better representations (HTS-pretrained CheMeleon, MolE 3D pretraining) that generalise across the structural divide.

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